

Assessing and modeling the contribution to Indoor Radon from the building materials: The S.I.R.E.M.[©] Model

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Abstract : - Accumulation of Indoor Radon activity concentration inside residential dwellings and workplaces is the result of a quite complex process which has many contributing sources (entry from soil, emission by building materials, from water and gas supplies). As it undergoes different physical mechanisms all occurring at the same time, thus, it turns out to depend on many parameters. They make the realization of a realistic and complete physico-mathematical model, that could be used on-site also by professionals, a really hard task to achieve.

Therefore, in the framework of an Interdisciplinary Strategic Programme of Applied Research for the Assessment of Naturally Occurring Radioactive Materials in Natural and Anthropogenic Environments, called RAD_CAMPANIA, carried out in Italy at the University of Salerno, by environmental physicists, geologists, environmental and building engineers, using an adaptive hierarchical multiscalar approach, an innovative procedure of assessing and modeling the contribution to Indoor Radon from the building materials: The S.I.R.E.M.[©] Model", has been realized. It has been applied a realistic case of a single-detached slab-on-grade house, made with concrete.

The procedure is simple and easy-to-use. It is entirely based on on-site experimental measurements of indoor Radon-in-air activity concentrations and Radon-in-air emissions by Building Materials, performed through the spectrometry of Radon short-lived alpha-emitting progenies. From the application of the S.I.R.E.M. procedure, interesting and useful indications can be derived about the dependence of the Indoor Radon values with the age of the concrete materials used for construction, the ventilation rate and the prediction about the goodness of the installation of engineering remedies to reduce the Indoor Radon levels.

Key-Words: – Radon emanation by building materials, Indoor Radon concentrations, energy efficiency

1 Introduction

Human exposure to Naturally Occurring Radioactivity (NOR) represents a large percentage of the total one to ionizing radiation from artificial and natural sources. In particular, among the natural sources, Radon (Radon-222) and its progeny, a radioactive gas produced during the decay chain of the primordial radionuclide Uranium-238, provides a contribution of almost 40%, in terms of world annual average value. Accordingly, since 1990 extensive epidemiological investigations have been performed and have demonstrated a statistically significant increase in the lung cancer risk from prolonged exposure to indoor Radon [1, 2]. Consequently, it has been established by the World Health Organization (WHO), that exposure to Radon, claimed by the International Agency for Research on Cancer (IARC) to be a carcinogenic substance of Group I, has to be considered as one of the major risk factors for lung cancer, when people are exposed to it inside confined spaces like residential buildings, workplaces and underground places like mines and caves [3].

Actually, the current legislation about the assessment of the population and workers exposure to natural ionizing radiation, due to Radon and its progeny, existing in the different European state members, fulfills the European Recommendation CEC 90/143/Euratom (1990) [4] and the Council Directive 96/29/Euratom (1996) [5]. The first one has only recommended some Reference Level (RL) values for dwellings (200 Bq/m³ and 400 Bq/m³ in new and old residential buildings, respectively) while, for workplaces, EC recommended Member States to establish an Action Level (AL) value within the range 500-1000 Bq/m³ [6]. Consequently, the different transposed national regulatory bodies require that, inside residential buildings and workplaces, the overall indoor Radon activity concentrations, averaged over a period of one year, only have to be lower than specific levels, without any consideration about the contribution of the different Radon sources (entry from underneath soil, emanation by building materials, entry from water and gas supplies), presence of artificial or natural ventilation and the geological features of the dwelling location, that could cause those values of concentration.

In particular, the most recent residential investigations show also that the risk increase can be relevant even at concentration levels of the order of 100 Bq/m³, lower than the level values prescribed previously by [4 -6] and transposed

into the different national legislations. These outcomes have been analyzed by a special committee of European experts, within the European project RADPAR [7], to put the basis of recommendations for a new comprehensive European Union radiation protection directive [8] that takes into account the population's and workers' exposure to both natural and artificial sources of radiation inside a unified framework of norms and prescriptions, indicating “... a new concept of exposure situations ...”, such that “... the provisions of Commission Recommendation 90/143/Euratom [4] are enabled to be incorporated into the binding requirements of the Basic Safety Standards while leaving enough flexibility for implementation ...”.

Thus, this European Council Directive 2013/59/Euratom of 5 December 2013, laying down basic safety standards for protection against the dangers arising from exposure to ionizing radiation, repealing Directives 89/618/Euratom, 90/641/Euratom, 96/29/Euratom, 97/43/Euratom and 2003/122/Euratom [8], gives more precise indications to the European Member States in matter of radioprotection from exposure to different sources of ionizing radiation. It poses, for the first time, in a harmonized framework, the issue of the human exposure to ionizing radiation in the most complete and general way, considering all the possible kind of sources (artificial and natural) and the different human targets (population and workers) in the same regulatory framework. As it has been underlined by F. Bochicchio [9] requirements for Radon protection for workers turn out to be are much more tightening than in the previous Recommendation [5, 6], while for the first time exposures to Radon in dwellings are regulated in a Directive.

In such a context it is also considered the role of the exposure due to indoor Radon and to the gamma radiation and Radon emanation by building materials, both of natural and artificial origin. The latter ones can be industrial products containing Naturally Occurring Radioactive Materials (NORM), eventually Technologically Enhanced (TE-NORM) in some cases. About that, the Directive states:

“... building materials emitting gamma radiation should be within the scope of this Directive but should also be regarded as construction products as defined in Regulation (EU) No 305/2011, in the sense that that Regulation applies to construction works emitting dangerous substances or dangerous radiation...”, and

“... Industries processing naturally-occurring radioactive material extracted from the earth's crust subject workers and, if material is released into the environment, members of the public to

increased exposure ...”, “ ... Protection against natural radiation sources, rather than being addressed separately in a specific title, should be fully integrated within the overall requirements. In particular, industries processing materials containing naturally-occurring radionuclides should be managed within the same regulatory framework as other practices ...”.

The unified approach of this Directive suggests that an interdisciplinary approach to the issues of the risk assessment associated to the exposure to ionizing radiations can be the right one to address such a complex issue and in the long term the only one really successful in achieving solutions and remedies to radioprotection of population and workers. A very accurate and detailed evaluation of the consequences of the issue of Directive 2013/59/Euratom with a particular concern to Radon exposure is given by F. Bochicchio, 2014 [9].

For what it concerns exposure to indoor Radon that means that the intertwined roles played by the different Radon sources (soil, building materials, etc.) together with ventilation and living uses cannot be neglected any longer. Therefore, it turns not anymore irrelevant that solutions and remedies to the health issues consequences to Radon exposure, that amounts to nearly the half of our total yearly exposure to natural ionizing radiation for the world population, cannot be achieved without involving necessarily different expertise and skills like building and industrial engineering, geology, environmental physics.

For such reasons an Interdisciplinary Strategic Programme of Applied Research for the Assessment of Naturally Occurring Radioactive Materials in Natural and Anthropogenic Environments, called RAD_CAMPANIA, with particular attention to Radon, has been started in Italy at the University of Salerno, in collaboration with the interUniversity Centre for Applied Research on the Prediction and Prevention of Major Hazards (C.U.G.R.I.), and the Regional Agency for the Environmental Protection of Campania (A.R.P.A. Campania), [10, 11]. The project is carried out by environmental physicists, geologists, environmental and building engineers, using an adaptive hierarchical multiscale approach to investigate several sub-projects: Soils, Waters and Buildings. The work that have been done so far (e.g. [10, 11]) shows that the environmental quality monitoring in workplaces or housings, can, no more, be only limited to integrated measurements, but must be necessarily capable to identify which is the main source (soil, building material, water or natural gas) responsible for the accumulation of indoor Radon to perform the most appropriate mitigation remedies (for an

existing building) and/or design new buildings in an innovative way in order to reduce the possible activity concentrations to the reference limits prescribed by the legislation and to be conformal also to the requirements of energy-efficient modern buildings, as it has been recently underlined by A. V. Vasilyev [12].

Regarding the contribution to the indoor Radon activity concentrations provided by the Building Materials its direct evaluation from the activity of the naturally occurring radionuclides contained in walls, ceilings and structural elements, which is expressed usually, both in the regulatory bodies and in the scientific literature, in terms of the activity concentration index (I) [13] or the radium equivalent activity (Ra-eq) [14], is extremely difficult if not impossible.

Therefore, for such reasons, in the framework of the Buildings sub-project a simplified approach, the Salerno Indoor Radon Emanation Model (S.I.R.E.M.) procedure, to assess indoor Radon activity concentrations based upon on-site experimental evaluations of Radon emanation by building materials, has been formulated with the aim to realize a tool not only with a scientific valence but also useful and precious for its applications to real situations [15, 16].

In this paper a synthetic description of its main ingredients together with a case-study real application are provided.

2 General Features

Radiation exposure inside buildings is mainly associated to the occurring of activity concentrations of Radon-222 (“Radon”) and Radon-220 (“Thoron”), together with their short-lived daughters (alpha-emitting radionuclides), accumulated in the room air from soil (almost the dominant contribution), building materials and water and gas supplies [17].

But, in order to evaluate the real overall ionizing radiation exposure for residents, it is correct to evaluate, in addition to this “internal” exposure, also the gamma ray (“external”) exposure from the activity concentrations of Naturally Occurring Radioactive Uranium-238, Thorium-232 and Potassium-40 and their decay products contained inside the walls, floors and ceilings.

In “modern” buildings where in many cases the use of concrete foundations isolates very efficiently the inside space from soil, which, therefore, contributes very little to Indoor Radon, the main contribution to the overall Indoor Radon activity concentration if not negligible, can be reasonably attributed to the Radon exhalation from the building materials. This assumption can be further strengthened if it is taken into account that investigations on the diffusion length of Radon in

different building materials have showed that in concrete the Radon diffusion length amounts to about 6 cm [18]. This means that a slab-on-grade foundation made of a concrete floor of 10 cm thickness “works” very well like a very efficient diffusion barrier for Radon from the soil. Only in case of cracks in the floor or in the absence of a concrete floor (for example, like in older dwellings in many rural areas of southern Italy) or with different typologies of foundations, a significant contribution to indoor Radon activity concentration may be expected especially in those areas with special geological structures and high concentrations of Uranium and Radium in the soil underneath the dwellings [10, 11].

In order to guarantee low levels of Radon activity concentrations inside a working or a living place, first of all it is necessary to be able to predict the maximum value of indoor Radon to which people can be exposed and which are the main sources causing it, and only then to be able to identify the most appropriate and cost-effective technique of mitigation that, when installed will guarantee the respect of the regulatory reference limits imposed by the national laws and will work efficiently to keep the occupants safe.

In order to do that, it is important to have a detailed knowledge of the Radon dynamics in a confined space. That means accurate predictions are possible in principle only with a very sophisticated and complete model of all the space and time-dependent physical processes involved. Scientific literature shows many examples of different attempts that have been proposed in the last 30 years. One of the most complete and accurate from the point of view of the physico-mathematical description of the problem have been realized by L. Font Guiteras in his RAGENA Model [19, 20]. But, in order to realize at the same time a procedure applicable to real cases preserving the reliability of the predictions and that could be pretty easily managed also by professionals, a steady-state model of prediction of Indoor Radon activity concentrations, christened S.I.R.E.M.[®] (Salerno Indoor Radon Emanation Model), has been developed in the framework of the RAD_Campania Model [10, 11]. The application of this innovative S.I.R.E.M. procedure to real cases enables not only the individuation of the major source of indoor Radon but also evaluation of the efficacy of the Radon protection system designed for the mitigation.

The particularity of the model relies on an entirely new approach, which is based on site experimental measurements with Radon-in-air analyzer, RAD7, manufactured by DURRIDGE Company, Inc. (Bedford, MA.). This provides a relevant simplification in the complexity of the

entire model and enables to distinguish clearly the different contributions of each source to the accumulation of Radon into a confined space.

In the following the main ingredients of the S.I.R.E.M. Model are briefly sketched and the model predictions are compared with the experimental results obtained in its application to a case study represented by a typical Italian detached house.

3 The Radon dynamics

Radon entry into buildings is caused by different sources, like the soil under or surrounding the buildings, the exhalation from building materials (BM), water and natural gas supplies. When a pressure gradient between indoor and outdoor occurs the outdoor air can also contribute to the overall indoor Radon level.

In the scientific literature it is widely accepted that the main source of Radon entry is the soil underneath and the BM [17]. Water and gas supplies contribute to indoor Radon levels only when they carry extremely high Radon activity concentrations like for example in the case of drinking water taken from groundwater wells and not from a distribution network made of aqueducts, where due to the different stages of capitation, potabilization and distribution, Radon levels turn out to be typically rather low due to the degassing mechanism [21]. Radon levels in the outdoor air, instead, are typically low due to the Radon dilution in the open atmosphere [17].

The whole process of Radon migration from the production location to its accumulation inside a confined space can be sketched in *four stages*: I. generation in the source medium (production rate depends on the radium content, grain size distribution, porosity and water content); II. migration inside the source medium and exhalation at the soil surface (through the physical mechanisms of diffusion – driven by Radon concentration gradients and ruled by the Fick’s law – and advection – driven by pressure gradients under the Darcy’s law); III. entry into a dwelling, depending, of course, on the typology of the interface between the source medium and the indoor space (material, presence of cracks and geometry); IV. Indoor Radon accumulation depending on the balance between the entry and removal processes (radioactive decay and ventilation mechanism).

At each stage it is easy to guess that many parameters of different origin take part into the process and most of them are time dependent. So, the evident complexity originated by the existence of this great number of parameters and processes has led to the development of partial models and experimental studies focused only on the

dynamics of one of the above four stages. Anyway, all the models developed to-date and the experimental studies have been very useful for understanding the basic processes that regulate the Radon dynamics. However, it can turn out to be very difficult to extrapolate these results to real inhabited houses because the models are based on a highly idealized representation of reality.

Among all the Radon entry models in the literature, the most interesting and complete is the RAGENA model [19, 20], whose approach leads to the development of a global description which collects the knowledge acquired from partial models. The requirements fulfilled by this global dynamic Radon model (GDRM) are the following ones: I. to take into account all the Radon sources, processes and parameters affecting indoor levels; II. to be adaptable to different inhabited houses, with the possibility of an easy changing of the parameters in order to simulate mitigation methods [20].

The RAGENA model is divided into several sectors describing the dynamics of each Radon source (soil, BM, water and gas) contribution to the entry process. Each sector includes one or more compartments, connected through inflows and outflows of Radon atoms. The mass balance, for each sector, between inflows and outflows gives a first order differential equation; the result is a set of coupled differential equations solved using the 4th order Runge-Kutta numerical method. Thus, the model's outputs will provide the time evolution of soil and the BM Radon concentration and the Radon level in each room.

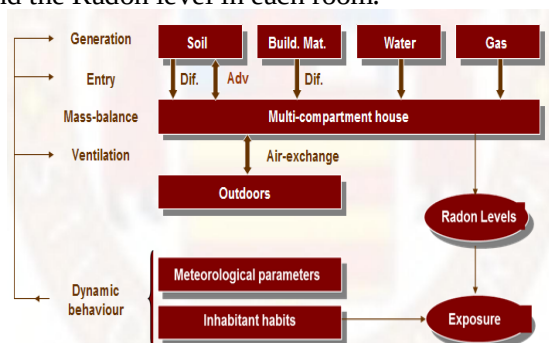


Figure 1- Structure of the RAGENA Model [20].

Starting from the RAGENA approach in 2006 Cocchioni and Giretti have developed a static model based on the construction of an expert system based on the Bayesian networks approach [22, 23]. This approach is widely applied in many branches of Civil Engineering [23]. If the task of the resolution procedure is only the prediction of a possible indoor maximum Radon concentration the static solution of the differential equations can be a valid approach as the error on the solution, in fact, turn out to be acceptable [23]. For each room the outputs of their approach consist in a range of

values of Radon levels and are aimed only to answering the simple question if the possible Radon indoor concentrations expose occupants to a high, medium or low Radon risk.

Processing these two approaches, starting from RAGENA model and using Cocchioni's assumptions and results, a first structure of a new predictive model have been built [15, 16], which has been modified, in a second time, to be adapted to the kind of experimental data processed by the RAD7 Radon detector.

4 The S.I.R.E.M.® Model

The global model sector developed christened S.I.R.E.M.® (Salerno Indoor Radon Emanation Model) has been developed in the framework of the software development environment STELLA Software (isee systems) designed for solving systems of ordinary differential equations.

In the model all the known sources are considered and each process has been modeled in a very simple way.

The S.I.R.E.M.® model is characterized to be:

1. evaluated in case of steady state conditions
2. semi-experimental
3. not site-specific.

The resolution of this sector model needs the measurements of the different emissions from all Radon sources and the knowledge of the soil parameters and the geometric characteristics of the room. The model is divided into a number of groups representing as many Radon sources (soil, building material, water, gas supply) as are present in the case study.

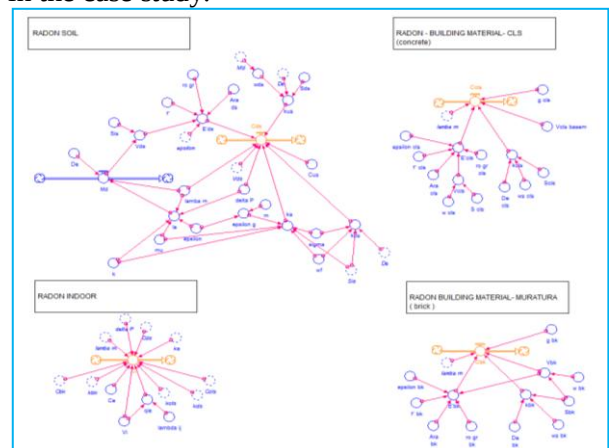


Figure 2- Structure of the S.I.R.E.M.® Model realized with the STELLA software development environment [15, 16].

5 Materials and methods

All the data necessary to run the model come from measurements obtained with the Radon-in-air analyzer, RAD7, whose heart is made of an

ion-implanted solid state detector for alpha emitting radionuclides, manufactured by DURRIDGE Company, Inc. (Bedford, MA.). RAD7 provides a continuous monitoring of the Radon-in-air activity concentrations (no matter if it is indoor air, soil-gas, air in thermal equilibrium with water rich of Radon, or air enriched of Radon emanated by building materials) through a spectral analysis of their alpha-emitting short-live progenies. This instrument has been chosen because of the following characteristics: (1) very low intrinsic background; (2) relatively high sensitivity in the “sniff mode” (^{218}Po counts) at $\sim 0.4 \text{ cpm/pCi/L}$ for ^{222}Rn in air; and (3) very good energy discrimination for the different Radon daughters [24]. In such a way the outcomes from the energy spectrum enable the user to easily distinguish between ^{222}Rn and ^{220}Rn just on the basis of the different alpha-energy signals recorded in the spectrum and produced by their progenies, like in a Crime Police Investigation different suspects can be distinguished by DNA analysis of their biological traces.

The set-up that has been made to measure on-site Radon emanation by Building Materials is made by a RAD7 unit in line connected to a 0,6 L Hard Surface Emission Chamber (HSEC), manufactured by DURRIDGE Company, Inc. (Bedford, MA.) [32], designed specifically for measuring Radon emanations from walls and floors directly on-site. The HSEC is connected through vinyl tubes to a air-drying unit, the Laboratory Drying Unit, containing DRIERITE (CaSO_4) plus 3% of CoCl_2 , working as moisture indicator, and to the RAD7 unit (Figure 3). Inside RAD7 an internal pump (1 L/min), draws air from the HSEC, through the dessiccant and an inlet filter, that unables the entry of Radon progeny into the monitor. The pump keeps the air circulating in a closed circuit mode. The filtered air decays inside the RAD7 unit, which consists of an electrostatic accumulation chamber and a ion implanted solid-state detector able to collect and count alpha radiation emitted by the short-lived alpha emitting Radon progeny, particularly the polonium isotopes. A high voltage of 2500 V is applied between the metal coated chamber walls and the solid state silicon detector, which converts alpha radiation directly to an electrical signal, discriminating the electrical pulses generated by α -particles from the different polonium isotopes (^{218}Po , ^{216}Po , ^{214}Po , ^{212}Po) with energies of 6.0, 6.7, 7.7 and 8.8 MeV, respectively. Using this approach, it is possible to use only the ^{218}Po peak for ^{222}Rn and ^{216}Po for ^{220}Rn (the so-called “sniff” mode), obtaining a rapid equilibrium between Polonium and Radon nuclides, because the equilibrium between ^{218}Po and ^{222}Rn is achieved in about 15 min (about five times the half-

life of ^{218}Po), and between ^{216}Po and ^{220}Rn in a few seconds.

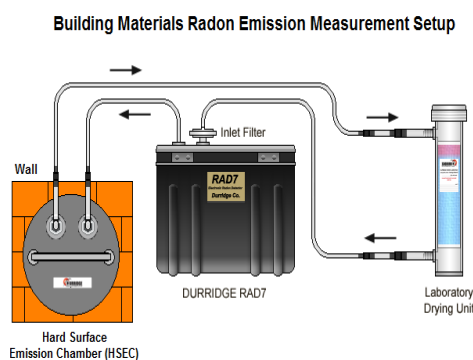


Figure 3 – Experimental layout for on-site measurements of Radon Emanation by Building Materials (courtesy by DURRIDGE Company, Inc. (Bedford, MA.)).

But, in order to achieve a full and satisfying applicability of the model a well planned measurement campaign must be necessarily organized.

To obtain a realistic value of Radon activity concentrations in soil, BM and indoor air a special user protocol, not included in the database of the RAD7 device, but developed as a consequence of many testing measurements performed in different matrices to find the the best settings. The final settings that has been chosen is the following: 2-hour runs for a continuous and simultaneous measurement of the activity concentrations in all the Radon sources: soil, indoor and BM.

Other important accuracies that have been taken into account concerns the choice of the most convenient locations for two classes of soil-gas measurements, both very important for the application of the model: unperturbed and perturbed soil-gas measurements, where perturbed soil means soil in the very close neighborhood of the building. As a reasonable assumptions distances of around 5 m from the external walls of the building have been considered as the limit beyond that the soil can be considered not influenced by the construction. If such a classification would not be possible then it will be necessary consider the maximum value of Radon activity concentration in soil-gas provided by Radon Prone Areas map of the territory [10, 11].

6 Application to a case – study (test-house) and preliminary results

In order to verify the validity of the above approach the S.I.R.E.M.© model has been applied to a test-house. The chosen test-house, entirely constructed with concrete, is a single-family detached house with a slab-on-grade foundation,

located in Baronissi, in the province of Salerno, Southern Italy.

The geology of the site is characterized by alluvial and volcanic deposit on a limestone base.

The house is built on a site with a medium-high level of risk on the Regional Radon map, that is characterized by a soil Radon concentrations from 20,000 to 50,000 Bq/m³ [10, 11].

Regarding the geometry the house has a rectangular plant on three levels: the basement with a garage of about 30 m², the groundfloor of about 150 m² and, finally, the area under the roof used as warehouse. The type of construction is typical of the Italian tradition of the 20th century: a framed concrete structure with external walls in concrete bricks.

The experimental set-up for Radon activity concentrations measurements consisting in the RAD7 unit equipped with HSEC for Radon emanation measurements from BM has been installed in the garage, which is principle the one more exposed to accumulation of indoor Radon, because closer to the main Radon entry source, the soil. This room is characterized by the absence of windows, just only a small opening on the front door of the garage, and the absence of a pavement. The floor in fact is made up of only a small slab of concrete. Two of the four perimetrical walls are in contact with the soil and so built totally in concrete because of their function is also that to contain the soil stress on them. The total volume of the test-structure is of about 100 m³.

How to proceed step-by-step

Step I: Individuation of all the Radon sources.

It is necessary to individuate all the possible Radon sources, taking part to the Indoor Radon accumulation process (Radon entry from disturbed- and undisturbed-soil, emanation from water and gas supplies, emanation from all the kinds of different building materials if present). As stated before for new buildings made with concrete (also for older buildings, provided that they are in general good conditions (absence of cracks, holes etc.)) like in the case the main source for indoor Radon accumulation is provided by the Radon emanation from the building materials. Therefore, this represents an ideal condition for applying the approach used by the SIREM model as the application can be based on accurate on-site experimental measurements of the emission by building materials (BM), differently from the application of all the other models from scientific literature, which rely on a huge number

of parameters difficult to be evaluated precisely and accurately at the same time.

In our case the Radon sources are represented by the building material and the soil (water and gas supply are not present) so the necessary measurements are: I. in the disturbed soil, II. Undisturbed soil, III. and by the only building material present: concrete.

“Disturbed soil” means the soil next to the external walls (inside 10 - 15 meters) while “undisturbed soil” has been considered the one that is located over those limits.

Step II. Choice of the measurement locations.

The ability of the operator to select and define the most convenient measurement stations in order to obtain the most representative Radon emission pattern from the different production sectors (Radon sources), to individuate the right open area fraction (e.g., the regions in the walls, ceiling and floor where the building materials shows defects cracks and other imperfections due to age and damages) and measure with accuracy the correct geometry of the site is fundamental in order that the application of the procedure will achieve the output which is closest to a realistic representation of the problem.

Step III. Choice of the data acquisition protocol.

Continuous measurements for Radon activity concentrations from soil and BM are performed with identical acquisition run protocols lasting from 30 minutes to 1 day, depending on the precision to achieve in the data.

Step IV. Performance of the measurements in the Radon production sectors (soil, BM) that have been individuated in **Step I**.

Following the above steps, the experimental results achieved together with the codes of the different measurement stations are summarized in Table 1 and graphically represented in Figure 4 below.

Step V. Introduction of all the required parameters into the model, see Table 2.

Some of them (labeled with code 1 in Table 2) can be measured on-site (as the room volume, the BM surface in contact with soil, the BM coverage, etc.), others (labeled with code 2 in Table 2) can be taken from numerous cases and experiments reported in the scientific literature (e.g., concrete efficacy diffusion coefficient, soil efficacy diffusion coefficient, etc.), others (like the pressure difference soil- external air) (labeled with 3) can be easily calculated with well-known basic equations.

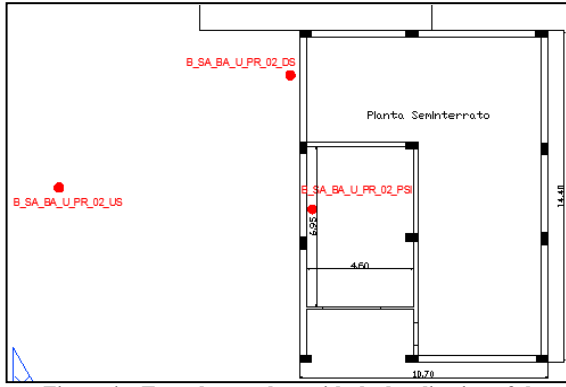


Figure 4 – Test - house plan with the localization of the measurement stations [15, 16].

STATION CODE	EXPERIMENTAL VALUE (kBqm ⁻³)
B_SA_BA_U_PR_02_DS	53.7
B_SA_BA_U_PR_02_US	51.3
B_SA_BA_U_PR_02_PS	0.12
B_SA_BA_U_PR_02_PS_bm	0.29

Table 1 - Radon activity concentrations from the different Radon production sectors measured outside and inside the test-house [15, 16].

PARAMETER	VALUE
V_i room volume	102.3 m ³ (1)
S_{cls} concrete surface	109 m ² (1)
$w_{s,cls}$ concrete coverage	0.0015 m (1)
λ_{rn} Radon decay constant	2.098·10 ⁻⁶ s ⁻¹ (2)
D_{cls} concrete eff. diffusion coefficient	5·10 ⁻⁸ m ² s ⁻¹ (2)
g_{cls} coverage factor	1 (2)
S_{is} surface in contact with soil	54.21 m ²
σ open area fraction	1x10 ⁻⁵ (2)
w_f foundations width	0.25m (1)
C_{ds} disturbed soil Rn concentration	53.7 kBqm ⁻³ (1)
C_{us} undisturbed soil Rn concentration	51.3 kBqm ⁻³ (1)
ε soil porosity	0.55 (3)
D_e soil eff. diffusion coefficient	10 ⁻⁷ m ² s ⁻¹ (2)
m water saturation fraction	0.32 (2)
C_{cls} concrete Radon emanation	0.29 kBqm ⁻³ (1)
ΔP pressure diff. soil-external air	0.85 Pa (3)
λ ventilation rate	0.4 h ⁻¹ (2)
C_e Radon-in-air concentration	0 Bqm ⁻³ (1)

Table 2 - Experimental data measured on-site and introduced in the SIREM© model [15, 16].

After performing the algorithms contained in the procedure the obtained calculated output (111 Bqm⁻³) turns out to be the maximum (steady-state) indoor Radon activity concentration that can be

reached in the actual structure condition when equilibrium is achieved.

As it can be noticed in Table 1 the calculated value (111 Bqm⁻³) is very close to the experimental value of the measured overall indoor Radon activity concentration (119 Bqm⁻³) on-site with the Radon detector RAD7. So, it confirms the validity of our approach, the goodness and the accurate construction of the S.I.R.E.M.© model and its equations describing the processes of indoor Radon accumulation.

Further, the outputs of the application of the model can be further elaborated also to predict the variability of Radon indoor activity concentration, if, for example, the structure conditions change due to the material ageing. In fact, it is possible to simulate this evolution changing the parameters regarding the building material like the open area fraction, Figure 5 or the ventilation of the room, Figure 6.

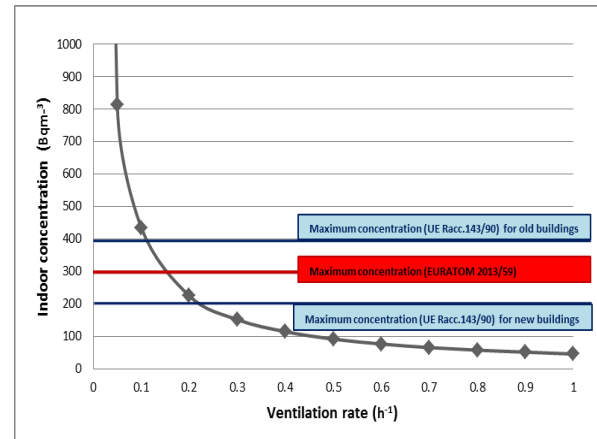


Figure 5 – Output of the simulation of the variability of indoor Radon activity concentrations as a function of the ventilation rate [15, 16].

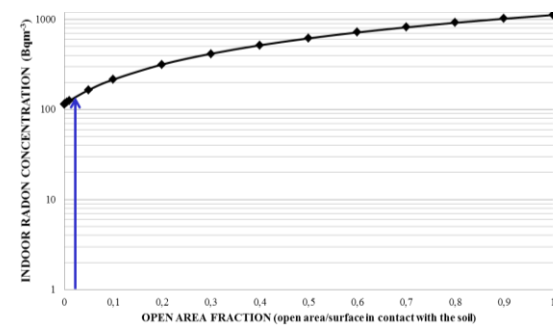


Figure 6 - Output of the simulation of the variability of the indoor Radon activity concentrations as a function of the open area fraction [15, 16].

In Figure 7 the output of the simulation of the behavior of the Indoor Radon activity concentrations versus the effective concrete diffusion coefficient, and, therefore, the concrete material age, is shown.

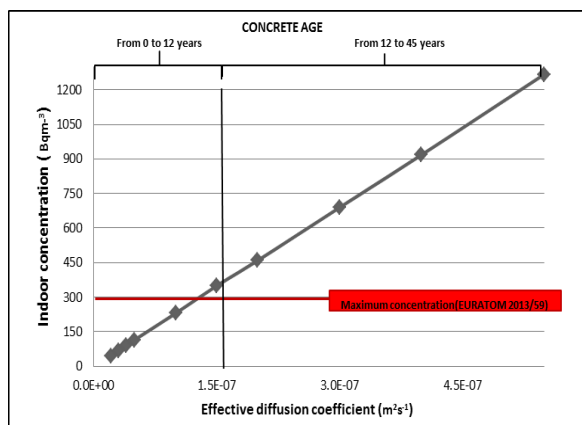


Figure 7- Influence of concrete age on indoor Radon activity concentrations [15, 16].

Finally, it has been evaluated also the efficacy, in reducing high Radon indoor activity concentrations, of recent Radon remediation techniques based on the installation of flexible-thin barriers (PETG, Polyethylene Naphtalate) that are simply mounted on the surface of the building materials, in direct contact with soil, in the case of fractured surface. The effective diffusion coefficient of the flexible - thin barriers are indicated in the several experimental cases reported in the scientific literature [25].

7 Concluding remarks

The latest European Council Directive 2013/59/Euratom of 5 December 2013 laying down basic safety standards for protection against the dangers arising from exposure to ionizing radiation, and repealing Directives 89/618/Euratom, 90/641/Euratom, 96/29/Euratom, 97/43/Euratom and 2003/122/Euratom gives accurate and severe indications to the European member states in matter of the exposure to natural ionizing radiation with particular attention to public's and workers' exposure to Radon. In order to accomplish the requirements that each state member legislation will indicate through national Radon plans it is not anymore possible to underestimate the roles and the interconnection played by the different Radon entry sources (soil, building materials, etc.) together with ventilation and dwellers' living habits. Also, it turns out to be more and more fundamental to design and realize buildings more and more complaints with the energy efficiency requirements [12]. Therefore, solutions and remedies to the health issues consequences to Radon exposure, that amounts to nearly the half of our total yearly exposure to natural ionizing radiation for the world population, cannot be achieved without a purely interdisciplinary approach in which different expertise and skills are necessarily involved

(building and industrial engineering, geology, environmental physics). In this interdisciplinary context a simple and easy-to-use procedure, the S.I.R.E.M. model, for the assessment of the different contribution to Indoor Radon accumulation in residential and workplaces has been implemented on the basis of experimental measurements that can be done directly on-site. The application to a test-site and the comparison, regarding the contribution from Building Materials, between the S.I.R.E.M. procedure predictions for the cumulative Indoor Radon activity concentration and the experimental results shows a very good agreement.

Besides that the S.I.R.E.M. results show also that it is possible to evaluate in quantitative way the role of the ventilation and of the age of the Building Materials (concrete in the examined case-study).

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